DO YOU KNOW...?

when oxidized one gram of fat gives 9 kcal.

10.12 Functions of fats

Fats perform several important functions:

1) Fats are a concentrated source of energy. one gram of fat gives 9 kilocalories of energy. Fat is stored in the body in the adipose tissue and provides energy to the body when required.
2) Fats are the constituents of cell membrane structure and regulate the membrane permeability.
3) Subcutaneous fat acts as an insulator and helps in retaining body heat.
4) They are essential for the digestion, absorption and utilization of fat soluble vitamins like Vitamin A, D, E and K.
5) Fats are important as cellular metabolic regulators (Steroid hormones and prostaglandin).
6) Fats have a sparing action on vitamin B_1 (i.e) if fat consumption is adequate, not much vitamin B_1 is needed.
7) Fats improve the palatability of the diet and give satiety value.(ie.)feeling of fullness in the stomach.
8) The calories in fat spare proteins from being oxidized for energy.
9) Cholesterol is needed for synthesis of sex and adrenal hormones (steroid hormones).
10) Substituting a fat high in PUFA or MUFA for a fat high in saturated fatty acids can decrease the level of blood cholesterol levels and hence reduce risk of heart disease.

10.12.1 Functions of Essential Fatty Acids

1) Maintenance of the functioning and integrity of cellular and sub-cellular membrane.
2) Regulation of cholesterol metabolism by transporting it between the blood and body tissues.
3) Acts as precursors for important group of hormone like compounds-prostaglandins which aid in regulating vascular function and help relieving pain and inflammation.
4) Delays blood clotting time.

10.13 Fat requirements

The ICMR recommended allowances of fat for Indians is given in **table 10.7.**

10.14 Deficiency and excess of fat in the diet

a) Effects of EFA deficiency

Deficiency of fat in the diet causes the deficiency of essential fatty acids. Deficiency of essential fatty acids leads to cessation of growth. It also results in flaky skin, development of itchy sores on the scalp. The common disorder in adults and children in India is phrynoderma or toad skin. The condition is characterized by the presence of horny eruptions on the posterior and lateral aspects of the limbs on the back and buttocks.

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